Akshat Tiwari

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Professional Summary

Computer Science undergraduate focused on building scalable systems, AI-powered products, and production-grade engineering solutions. Experienced in backend development, machine learning, deep learning, and full-stack engineering with hands-on work in Spring Boot, FastAPI, ReactJS, Docker, PostgreSQL, and cloud-native workflows.

Strong foundation in data structures, algorithms, system design fundamentals, and software engineering principles with active open-source contributions to Apache and Delta Lake ecosystem projects. Passionate about building real-world applications with high ownership, scalability, and strong engineering practices.

Education

Shri Ramdeobaba College of Engineering and Management — Nagpur, India
Bachelor of Technology in Computer Science and Engineering — *2023 – 2027*

Technical Skills

Languages: Java, Python, C, C++
Frameworks & Libraries: Spring Boot, FastAPI, ReactJS, Pandas, NumPy, Scikit-Learn, Tailwind CSS
Databases & Tools: PostgreSQL, MongoDB, Redis, Docker, Git, Linux, Kafka
Core Concepts: Data Structures & Algorithms, REST APIs, OOP, DBMS, Operating Systems, Machine Learning

Projects

Insurance Claim Processing System | GitHub — Apr 2026

- Designed and developed an enterprise-grade insurance claim processing platform using microservices architecture and Spring Boot.
- Implemented secure claim submission, policy management, document verification, and multi-level approval workflows with JWT-based authentication.
- Built scalable REST APIs and asynchronous processing pipelines for claim validation, fraud detection, and real-time notification handling.
- Integrated Docker-based containerization, API Gateway, and service discovery architecture for scalable deployment and system reliability.
- Designed role-based workflows for customers, surveyors, agents, and administrators with audit logging and centralized monitoring support.

HVAC Complaint Intelligence System | GitHub — May 2026

- Built an AI-powered complaint intelligence platform capable of clustering and analyzing large-scale HVAC customer complaints in real time.
- Reduced issue discovery timelines from weeks to hours using sentence-transformers, Gemini API, and HDBSCAN-based unsupervised learning workflows.
- Engineered scalable REST APIs with FastAPI and containerized deployments using Docker to ensure production-grade scalability, reliability, and maintainability.
- Developed an analytics dashboard to identify hidden service failures, anomaly trends, and operational bottlenecks.

Lung Severity Detection | GitHub — Mar 2026

- Built a deep learning-based medical imaging system for automated lung severity assessment using CT scans and X-ray datasets.
- Designed end-to-end ML workflows including preprocessing, model training, evaluation, and prediction automation pipelines.
- Applied advanced computer vision and image analysis techniques to improve diagnostic consistency and severity prediction accuracy.
- Created a scalable AI-assisted workflow to support faster and more accurate clinical decision-making processes.

Open Source Contributions

- Contributed to **apache/gravitino** across Java, Python, and JavaScript ecosystems to improve data catalog infrastructure and developer tooling.
- Contributed to **delta-io/delta-kernel-rs** using Rust and C for improving native Delta Lake interoperability and storage integration systems.
- Actively collaborate through pull requests, debugging, feature implementations, and issue discussions across open-source engineering communities.

Achievements

- Solved 450+ Data Structures and Algorithms problems across coding platforms demonstrating strong problem-solving abilities.
- Participated in 8+ hackathons with finalist positions in 2 competitions, building scalable engineering solutions under pressure.
- Successfully completed Hacktoberfest 2024 through multiple open-source contributions on GitHub.